

1. Overview for AudioPeakRMSChecker.ps1

This script analyzes the audio tracks inside a movie file (MKV, MP4). It checks for:

Peak (dB) loudest moment in the audio.

RMS (dB) average loudness over time.

Crest Factor (dB) difference between Peak and RMS

Clipping (number of samples that hit or exceed 0 dB)

The function is to give you a quick PASS/FAIL result so you can tell whether the audio is clean, safe, and ready for encoding, downmixing, or playback. It supports common multichannel formats including TrueHD, AAC, EAC3/DDP.

This tool does **not** modify audio. It only analyzes it and reports the results.

2. Requirements

To run AudioPeakRMSChecker, you need:

- PowerShell 7.6
- FFmpeg 8.1 (FFmpeg and FFprobe must be available on PATH or placed next to the script.)

Cross-Platform Compatibility:

- Windows 11
- Linux (Ubuntu, Debian, Arch, Fedora)
- macOS (Ventura, Sonoma)

As long as PowerShell 7.6 is installed and FFmpeg 8.1 binaries are placed next to the script, it works identically across all platforms.

Official FFmpeg download page: <https://ffmpeg.org/download.html>

To run the script:

```
pwsh -ExecutionPolicy Bypass -File .\AudioPeakRMSChecker.ps1 ".\YourMovie.mkv"
```

 The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

3. Supported Audio Formats

AudioPeakRMSChecker analyzes the following audio formats:

Codec Family	Formats Included
TrueHD	TrueHD 7.1
AAC	AAC 7.1
EAC3 / DDP	Dolby Digital Plus 5.1
DTS	DTS-HD 7.1

The tool automatically detects:

- the codec
- the number of channels
- whether the track is a source (7.1) or downmix (5.1)
- whether the track is supported for analysis

Unsupported tracks are ignored safely.

4. How the Analysis Works

The script performs the following steps:

1. Scans the movie file

It looks for audio tracks and identifies their codec and channel layout.

2. Runs FFmpeg's "astats" filter

This filter measures loudness, peaks, clipping, and dynamic range.

3. Collects the results

The script extracts the important numbers from FFmpeg's output.

4. Checks for problems

It looks for:

- peaks too close to 0 dB
- audio that is too loud overall
- low dynamic range
- clipped samples

5. Returns PASS or FAIL

PASS means the audio is clean and safe.

FAIL means the audio may distort or has loudness issues.

6. Saves a log file (*.astats.txt)

A text file is created with the full FFmpeg analysis for reference.

The entire process is automatic and requires no technical knowledge.

5. What Each Metric Means

AudioPeakRMSChecker reports four main metrics:

Peak (dB)

- The loudest moment in the audio.
- Values **near 0 dB** mean the audio is very loud.
- Values **above 0 dB** indicate clipping.

RMS (dB)

- The average loudness over time.
- Higher RMS (closer to 0 dB) means the audio is louder.
- Lower RMS (−20 dB to −35 dB) means more dynamic range.

Crest Factor (dB)

- The difference between Peak and RMS.
- Higher crest = more dynamic, cinematic sound.
- Low crest = compressed or flat audio.

Clipped Samples

- The number of samples that hit or exceed 0 dB.
- A few clips are usually harmless.
- Many clips indicate distortion.

These metrics help determine whether the audio is clean or risky.

6. PASS / FAIL Rules (Simple)

The script marks a track as **FAIL** if any of the following are true:

- **Peak > 0 dB**

The audio is clipping.

- **RMS louder than −12 dB**

The audio is extremely loud and may distort.

- **Crest < 6 dB**

The audio is heavily compressed.

- **Clipped samples > 5**

Too many clipped peaks.

- **Peak ≥ 0 dB with any clipped samples**

Even a single clipped sample triggers FAIL when the peak is at or above 0 dB.

If none of these conditions are met, the track is marked **PASS**.

PASS means:

- clean audio
- safe headroom
- healthy dynamics
- no distortion

FAIL means:

- the audio may distort

- the mix may be too loud
- the track may need correction

7. JSON Output Mode

AudioPeakRMSChecker can output results in **JSON format**, which is useful for:

- automation
- pipelines
- dashboards
- batch processing

Located at top of script. Line32

To **enable** JSON mode: \$JsonMode = \$true

To **disable** JSON mode: \$JsonMode = \$false

In JSON mode:

- No summary is printed
- No color output
- No interpretation text
- Only the JSON object is returned

Exit Codes:

number	description
0	Success
1	Input validation: file missing / no audio streams / no supported codecs / no valid 7.1 or DDP 5.1 combination
2	FFmpeg produced no stderr output (analysis aborted)
3	FFmpeg exited with non-zero exit code
5	FFmpeg exceeded maximum allowed runtime (timeout guardrail)
6	astats output missing required metrics (Peak / RMS / Peak count)
7	Peak level dB failed numeric parse
8	RMS level dB failed numeric parse
9	Peak count failed numeric parse
10	astats log file was not created on disk
11	astats log file is empty or smaller than 10 bytes
12	ffmpeg or ffprobe not found on PATH